

Time: 3 Hrs.

CTA

Max. Marks: 180

PAPER 1

JEE – ADVANCE – 2020 – P1 – Model

IMPORTANT INSTRUCTIONS

COMPLETE SYLLABUS-XI

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total Marks
Sec – I(Q.N : 1 – 6)	Questions with Single Answer Type	+3	-1	6	18
Sec – II(Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec – II(Q.N : 13 – 18)	Questions with Numerical Answer Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30 Decimal Numbers)	+3	0	6	18
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total Marks
Sec – I(Q.N : 19 – 24)	Questions with Single Answer Type	+3	-1	6	18
Sec – II(Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec – II(Q.N : 31 – 36)	Questions with Numerical Answer Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30 Decimal Numbers)	+3	0	6	18
Total				18	60

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total Marks
Sec – I(Q.N : 37 – 42)	Questions with Single Answer Type	+3	-1	6	18
Sec – II(Q.N : 43– 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec – II(Q.N : 49– 54)	Questions with Numerical Answer Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30 Decimal Numbers)	+3	0	6	18
Total				18	60

Exam Syllabus

PHYSICS	COMPLETE 11TH CLASS SYLLABUS	ONLY 😊
CHEMISTRY	COMPLETE 11TH CLASS SYLLABUS	
MATHEMATICS	COMPLETE 11TH CLASS SYLLABUS	

PHYSICS

Max Marks: 60

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

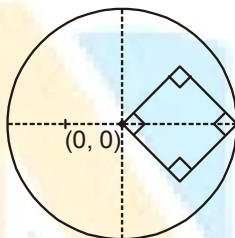
This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. A stone is projected vertically upwards from an elevated point A . When the stone reaches a distance h below A , its velocity is double of what it was at a height h above. The greatest height attained by the stone with reference to point A is

(A) $2h$ (B) $3h$ (C) $\frac{5h}{3}$ (D) $\frac{2h}{3}$

2. A square hole is punched out of a circular lamina, the diagonal of the square being the radius of the circle. If D is the diameter of the circle. The distance of C.G. of the remainder from the centre of the circle is



(A) $-\frac{D}{2(2\pi-1)}$ (B) $-\frac{D}{4(2\pi-1)}$ (C) $-\frac{D}{2(4\pi-2)}$ (D) $-\frac{D}{4(4\pi-2)}$

3. How far does a wooden (spherical) ball of specific gravity 0.4 and radius 2 feet sink in water?

(A) 1.25 feet (B) 1.75 feet (C) 0.75 feet (D) 0.50 feet

4. A vessel from which the air is rapidly being pumped out contains a small amount of water at 0°C . The intense evaporation causes a gradual freezing of the water. What fraction of the original water can be converted into ice by this method? (Latent heats of vaporisation and fusion are $L_V = 539 \text{ cal/g}$ and $L_f = 80 \text{ cal/g}$ respectively)

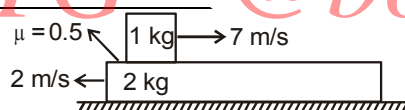
(A) 0.63 (B) 0.36 (C) 0.87 (D) 0.45

5. Position of a particle as measured from mean position at the end of three successive second is 1 m, 5 m, 5 m. Determine the time period of oscillation.

(A) $T = \frac{2\pi}{\cos^{-1}\left(\frac{4}{5}\right)}$ (B) $T = \frac{2\pi}{\sin^{-1}\left(\frac{4}{5}\right)}$ (C) $T = \frac{2\pi}{\cos^{-1}\left(\frac{3}{5}\right)}$ (D) $T = \frac{2\pi}{\sin^{-1}\left(\frac{3}{5}\right)}$

6. In the figure two blocks are given velocities as shown

Rough Work



The lower block is very long. Find out the time, when friction force between two blocks will become zero

- (A) 0.6 s (B) 1.2 s (C) 2.4 s (D) Will never become zero

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks.

7. Molar heat capacity of an ideal gas varies as $C = C_V + \alpha T$, $C = C_V + \beta V$ and $C = C_V + \gamma P$, where α , β and γ are constants.
 - (A) Equation of process in case of $C = C_V + \alpha T$ is $Ve^{-\alpha T/R} = \text{constant}$
 - (B) Equation of process in case of $C = C_V + \beta T$ is $Te^{R/\beta V} = \text{constant}$
 - (C) Equation of process in case of $C = C_V + \gamma P$ is $V = \gamma nT$
 - (D) Equation of process in case of $C = C_V + \gamma P$ is $Ve^{R/\beta P}$
8. The air column in a pipe closed at one end is made to vibrate in its second overtone by a tuning fork of frequency 440 Hz. The speed of the sound in air is 330 m/s. End corrections may be neglected. Let P_0 denote the mean pressure at any point in the pipe and ΔP_0 the maximum amplitude of pressure variation. Choose the correct statements.
 - (A) Pressure amplitude at the middle is $\frac{\Delta P_0}{\sqrt{2}}$
 - (B) Maximum pressure at the open end of the pipe is P_0
 - (C) Minimum pressure at the open end of the pipe is P_0
 - (D) Maximum and minimum pressure at the closed end of the pipe is $P_{\max} = P_0 + \Delta P_0$ and $P_{\min} = P_0 - \Delta P_0$
9. The block A of mass m connected with a spring of force constant K is executing SHM. The displacement-time equation of the block is $x = x_0 + A \sin \omega t$. An identical block B moving towards negative x -axis with velocity V_0 collides elastically with block A at time $t = 0$. Then
 - (A) After collision amplitude will be $V_0 \sqrt{\frac{m}{K}}$
 - (B) Velocity of B just after collision will be $a\omega$ towards positive x -direction

Rough Work

(C) Displacement time equation of A after collision will be $x = x_0 - V_0 \sqrt{\frac{m}{K}} \sin \omega t$

(D) Displacement time equation of A after collision will be $x = x_0 + V_0 \sqrt{\frac{m}{K}} \sin \omega t$

10. Which of the following functions of x and t represent a progressive wave?

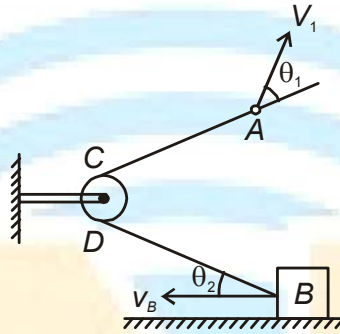
(A) $y = \sin (4t - 3x)$

(B) $y = \cos (4t + 3x)$

(C) $y = \frac{1}{3 + (2t - x)^2}$

(D) $y = \frac{1}{4t + 3x}$

11. In the given figure, block is connected by an inextensible string. Choose the correct statements.



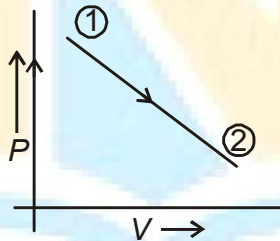
(A) $v_1 \cos \theta_1 = v_B \cos \theta_2$

(B) $v_1 \cos \theta_1 = v_B \sin \theta_2$

(C) $\omega_{A/C} = \frac{v_1 \sin \theta_1}{AC}$ and $\omega_{B/D} = \frac{v_B \sin \theta_2}{BD}$

(D) $\omega_{A/C} = \frac{v_1 \cos \theta_1}{AC}$ and $\omega_{B/D} = \frac{v_B \cos \theta_2}{BD}$

12. P - V curve is shown in figure for ideal gas, with negative slope and positive intercept $P = -aV + b$



(A) During expansion the temperature of the gas first increases, reaches a maximum and then decreases

(B) Maximum temperature is $T_{\max} = \frac{b^2}{4nRa}$

(C) During expansion the temperature of the gas first decreases, reaches a minimum and then increases

(D) Minimum temperature is $T_{\min} = \frac{b^2}{nRa}$

Rough Work

SECTION-III

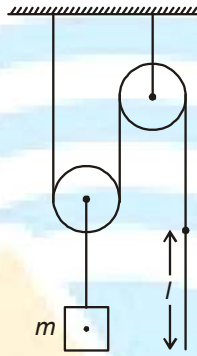
(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths Place).

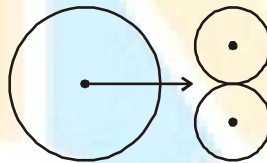
eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme: +3 for correct answer, 0 in all other cases.

13. In the figure shown, friction force between the bead and the light string is $\frac{mg}{4}$. The time in which the bead loose contact with the string after the system is released from rest is $t = \sqrt{\frac{nl}{7g}}$. Then find n?



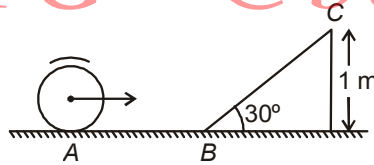
14. Two equal discs initially at rest are in contact on a table. A third disc of same mass but of double radius strikes them symmetrically and itself comes to rest after impact. The coefficient of restitution is $\frac{n}{3}$ then find n ?



15. A particle of mass 1 kg is projected with velocity $20\sqrt{2}$ m/s at an angle 45° from ground. When the particle is at highest point ($g = 10$ m/s²), angular velocity of the particle about point of projection is $\frac{n}{10}$ then rad /s

16. A small sphere of mass 1 kg is rolling without slipping with linear speed $v = \sqrt{\frac{200}{7}}$ m/s. It leaves the inclined plane at point C. Linear speed at point C is $v = \sqrt{\frac{100}{n}}$ m/s, assume inclined plane to be fixed and friction to be sufficient for pure **rolling** also there is no loss of energy during impact between sphere and inclined plane.

Rough Work



17. A bird moves with velocity 20 m/s in a direction making an angle of 60° with the eastern line (take as x -axis) and 60° with vertical upward (take as z -axis). The velocity vector in rectangular form is $n \times 10 \left[\frac{1}{2}i + \frac{1}{\sqrt{2}}j + \frac{1}{2}k \right]$ then find n ?
18. A rod has a total charge Q uniformly distributed along its length L . If the rod rotates with angular velocity ω about its end. Its magnetic moment is given by $\frac{Q\omega L^x}{Y}$. The value of $\frac{Y}{x}$ is ?



THE NARAYANA GROUP

Rough Work

SECTION – I

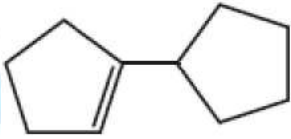
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

19. In an inert gas, the number of d, s and p electrons are x, y and z respectively. If $x = z - y$, then select the correct statement about inert gas.

(A) Period number of inert gas is 5
(B) Period number of inert gas is 4
(C) $x = 20$
(D) $y = 10$

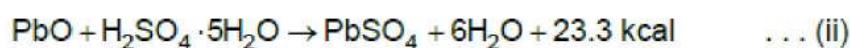
20. Major product formed when  is treated with dilute sulphuric acid is



21. 0.96 g of HI were heated to attain equilibrium $2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$. The reaction mixture on titration requires 15.7 mL of N/10 hypo solution. Calculate degree of dissociation of HI.

(A) 0.418
(B) 0.209
(C) 0.627
(D) 0.836

22. The ΔH_f for SO_3 will be



(A) 101.2 kcal
(B) – 101.2 kcal
(C) 146.8 kcal
(D) – 146.8 kcal

23. Consider the given statements about hydrides.

S_1 : Density of alkali metal hydrides is greater than that of the metal from which they are formed. This is explained by H^- occupying holes in the lattice of the metal without distorting the metal lattice.

S_2 : The metallic hydrides are less dense than the parent metal because of expansion of crystal lattice through the inclusion of hydrogen.

Rough Work

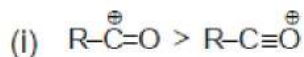


Select the correct statement.

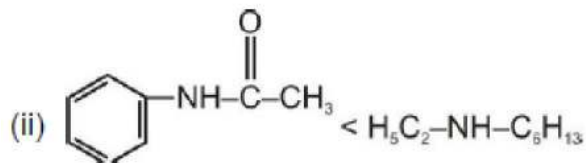
- (A) S_1 is true and S_2 is false
(C) Both S_1 and S_2 are false

- (B) S_1 is false and S_2 is true
(D) Both S_1 and S_2 are true

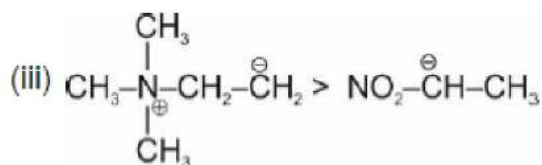
24. Use T for correct information and F for False about mentioned characteristic in bracket.



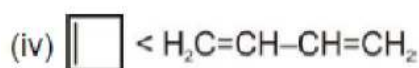
(Stability)



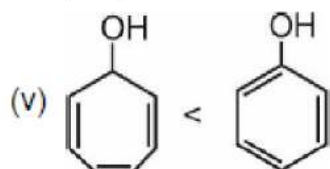
(C-N bond angle)



(Stability of Carbanion)



(Heat of combustion) (magnitude)



(Acidic nature)

- (A) TFTFT
(C) FTFTF

- (B) FFFFT
(D) FFTFT

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks.

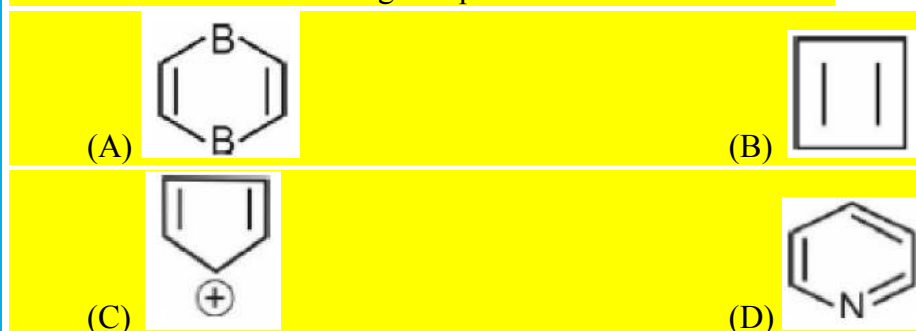
25. Which of the following statements is/are correct?

- (A) $LiH > NaH > KH > RbH > CsH$ (Thermal stability)
(B) KO_2 is paramagnetic in nature
(C) Milk of magnesia is used as antacid
(D) $BeCl_2$ has polymeric structure in solid state

Rough Work

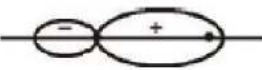


26. Which of the following compound is aromatic in nature?



27. Which of the following is/are correct regarding MOT?

(A) $\psi_A + \psi_B$ gives bonding molecular orbital.

(B)  is bonding molecular orbital

(C) The probability density of finding the electrons in bonding molecular orbital equal to $\psi_A^2 + \psi_B^2 + 2\psi_A\psi_B$

(D) Electron density increases between the nuclei of bonding M.O.

28. Which of the following reactions is/are disproportionation reactions?

(A) $Cl_2 + NaOH \xrightarrow{\Delta} NaCl + NaClO_3 + H_2O$ (B) $NH_4NO \xrightarrow{\Delta} N_2 + H_2O$

(C) $NH_4NO_3 \xrightarrow{\Delta} N_2O + H_2O$ (D) $H_3PO_3 \xrightarrow{\Delta} H_3PO_4 + PH_3$

29. The solubility of a sparingly soluble salt A_xB_y in water at $25^\circ C$ is $1.4 \times 10^{-4} M$. The solubility product is 1.1×10^{-11} . The possibilities are:

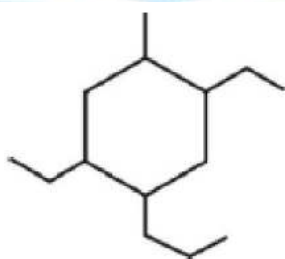
(A) $x = 1, y = 2$

(B) $x = 2, y = 1$

(C) $x = 1, y = 3$

(D) $x = 3, y = 1$

30. Identify the correct statement(s) for the following molecule:



(A) IUPAC name of the given molecule is 1, 4-Diethyl-2-methyl-5-propylcyclohexane

(B) IUPAC name of the given molecule is 2, 5-Diethyl-4-methyl-1-propylcyclohexane

(C) Number of 3° carbon atoms in given molecule is four

(D) Parent chain in the above molecule is open chain

Rough Work

SECTION-III

(Numerical Value Answer Type)

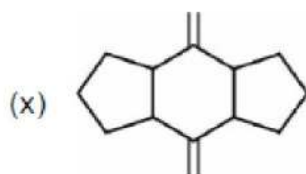
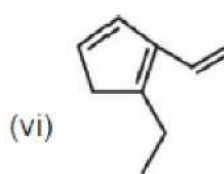
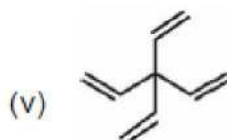
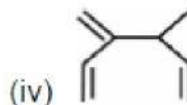
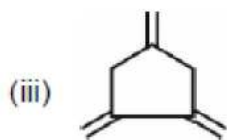
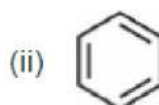
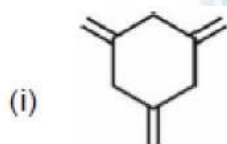
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eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme: +3 for correct answer, 0 in all other cases.

31. The internal energy change in the conversion of 1.0 mole of the calcite form of CaCO_3 to the aragonite form is +0.21 kJ. Calculate the enthalpy change when the pressure is 1.0 bar, given that the densities of the solids are 2.71 g cm^{-3} and 2.93 g cm^{-3} respectively. Express your answer as $(\Delta H/35)$ in J mol^{-1} .

32. How many of the following compounds have same general formula $\text{C}_n\text{H}_{2n-6}$?

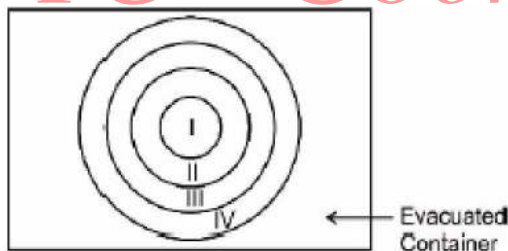


33. 2 mole of ideal gas undergoes a state change from 2 atm to 20 atm at constant temperature of 300 K.

What is the change in Gibb's free energy for this state change in kJ? (Use $R = 8 \text{ J K}^{-1} \text{ mol}^{-1}$ and $\ln 10 = 2.3$)

34. The inner sphere 'I' of a vessel contains CH_4 at pressure 4 atm and Helium at pressure 2 atm. Gases are allowed to effuse through series of spherical vessels through fine holes as shown in figure. The ratio of rate of effusion of He to CH_4 when they are finally released to an evacuated container is $X : 1$. The value of X is

Rough Work



35. pH of 0.1 M NaHCO_3 (given : pK_{a1} and pK_{a2} of H_2CO_3 is 7 and 10) is X. The value of $10X$ is
36. A solution of H_2O_2 is diluted to 10 times of its initial volume. 20 mL of this diluted solution requires 10 mL of 0.2 N KMnO_4 for complete oxidation. The volume strength of original H_2O_2 solution is V. The value of $10V$ is

MATHEMATICS

Max Marks: 60

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

37. $\sum_{k=0}^{20} \binom{20}{k}^2$ is equal to:
 (A) $^{40}C_{21}$ (B) $^{40}C_{19}$
 (C) $^{40}C_{20}$ (D) $^{41}C_{20}$
38. If $|z - 2 - i| = |z| \left| \sin \left(\frac{\pi}{4} - \arg z \right) \right|$, then locus of z is
 (A) a pair of straight lines (B) circle
 (C) parabola (D) ellipse
39. Let $P(6, 3)$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal at the point P intersects the x -axis at $(9, 0)$, then the eccentricity of the hyperbola is
 (A) $\sqrt{5/2}$ (B) $\sqrt{3/2}$
 (C) $\sqrt{2}$ (D) $\sqrt{2}$
40. If m is the A.M. of two distinct real numbers ℓ and n ($n > \ell$) and G_1, G_2 and G_3 are three geometric means between ℓ and n , then $(G_1)^4 + 2(G_2)^4 + (G_3)^4$ equals
 (A) $4 \ell^2 mn$ (B) $4/m^2 n$
 (C) $4 \ell mn^2$ (D) $4 \ell^2 m^2 n^2$

Rough Work

41. Let α and β be such that $\pi < \alpha - \beta < 3\pi$. If $\sin \alpha + \sin \beta = -\frac{21}{65}$ and $\cos \alpha + \cos \beta = -\frac{27}{65}$, then the value of

$$\cos \frac{\alpha - \beta}{2} \text{ is}$$

(A) $-\frac{3}{\sqrt{130}}$

(B) $\frac{3}{\sqrt{130}}$

(C) $\frac{6}{65}$

(D) $-\frac{6}{65}$

42. In triangle ABC, the equation of side BC is $x - y = 0$. The circumcenter and orthocenter of the triangle are (2, 3) and (5, 8), respectively. The equation of the circumcircle of the triangle is

(A) $x^2 + y^2 - 4x + 6y - 27 = 0$

(B) $x^2 + y^2 - 4x - 6y - 27 = 0$

(C) $x^2 + y^2 + 4x + 6y - 27 = 0$

(D) $x^2 + y^2 + 4x + 6y - 27 = 0$

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks.

43. The equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ is/are

(A) $\sqrt{3}y = 3x + 1$

(B) $\sqrt{3}y = -(x + 3)$

(C) $\sqrt{3}y = x + 3$

(D) $\sqrt{3}y = -(3x + 1)$

Rough Work



44. If a, b, c are in G.P., then the roots of the equation $ax^2 + bx + c = 0$ are in the ratio

(A) $\frac{1}{2}(-1 + i\sqrt{3})$

(B) $\frac{1}{2}(1 - i\sqrt{3})$

(C) $\frac{1}{2}(-1 - i\sqrt{3})$

(D) $\frac{1}{2}(1 + i\sqrt{3})$

45. The equation of a circle of radius 1 touching the circles $x^2 + y^2 - 2|x| = 0$ is

(A) $x^2 + y^2 + 2\sqrt{2}x + 1 = 0$

(B) $x^2 + y^2 - 2\sqrt{3}y + 2 = 0$

(C) $x^2 + y^2 + 2\sqrt{3}y + 2 = 0$

(D) $x^2 + y^2 - 2\sqrt{2}y + 1 = 0$

46. The graph of the quadratic trinomial $y = ax^2 + bx + c$ has its vertex at $(4, -5)$ and two x -intercepts, one positive and one negative. Which of the following holds good?

(A) $a > 0$

(B) $b < 0$

(C) $c < 0$

(D) $8a = b$

47. The value of ${}^nC_1 + {}^{n+1}C_2 + {}^{n+2}C_3 + \dots + {}^{n+m-1}C_m$ is equal to

(A) ${}^{m+n}C_n - 1$

(B) ${}^{m+n}C_{n-1}$

(C) ${}^mC_1 + {}^{m+1}C_2 + {}^{m+2}C_3 + \dots + {}^{m+n-1}C_n$

(D) ${}^{m+n}C_m - 1$

48. If $\sin^3 \theta + \sin \theta \cos \theta + \cos^3 \theta = 1$, then θ is equal to ($n \in \mathbb{Z}$)

(A) $2n\pi$

(B) $2n\pi + \frac{\pi}{2}$

(C) $2n\pi - \frac{\pi}{2}$

(D) $n\pi$

SECTION-III

(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths Place).

eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme: +3 for correct answer, 0 in all other cases.

49. The greatest value of x^2y^3 where x and y lie in the first quadrant on the line $3x + 4y = 5$ is $\frac{\lambda}{32}$ then the value of λ is

50. If θ_1 and θ_2 are two values lying in $[0, 2\pi]$ for which $\tan \theta = \lambda$, then $\left| \frac{\tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2}}{2} \right|$ is equal to

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51. For any integer k , let $\alpha_k = \cos \frac{k\pi}{7} + i \sin \frac{k\pi}{7}$, where $i = \sqrt{-1}$. Value of the expression

$$\frac{1}{8} \frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^3 |\alpha_{4k-1} - \alpha_{4k-2}|} \text{ is}$$

52. The absolute value of the sum of the abscissas of all the points on the line $x + y = 4$ that lie at a unit distance from the line $4x + 3y - 10 = 0$ is $\frac{\lambda}{5}$ then the value of λ is

53. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$, then the maximum value of $\frac{1}{8}ab$ is

54. If a , b and c are in G.P. and x , y , respectively, are the arithmetic means between a , b and b , c , then the value of $\frac{1}{5} \left(\frac{a}{x} + \frac{c}{y} \right)$ is



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